

A concavity obstruction and repaired support reductions in a Frankl entropy programme

Audit of Liu’s arXiv v1 — unrefereed candidate; global inequality open

Anonymous

14 August 2026

Contents

Abstract	1
1. Frozen model and the unresolved obligation	3
2. A parameterised positive-curvature direction in the arXiv-v1 proof domain	4
3. Repaired cardinality reductions	6
4. PSD argument, exact finite-bound certificate, and trust boundary	9
5. Entropy-zero identity, sharp asymptotic, and ceiling	11
6. Certified one-component atom-zero face	13
7. Exact elimination of q : ten generic variables and a six-variable subface	14
8. Assurance boundary and artifact map	15
9. Limitations, availability, and declarations	18
References	18

Abstract

Release status: UNREFEREED CANDIDATE / GLOBAL INEQUALITY NOT PROVED.

This note studies an entropy programme for Frankl’s union-closed sets conjecture at the frozen exploratory target $c = 153/400 = 0.3825$. It does **not** prove that this constant is valid and establishes **no new lower bound** for Frankl’s conjecture. Its principal result is instead an explicit admissible direction of positive curvature for every protocol parameter $0 < \ell \leq 1$. Under an exact source-to-model map, this invalidates the joint-concavity inference printed in the proof of Theorem 12 of arXiv:2306.08824v1, including the theorem’s “sufficiently small ℓ ” regime. This is an obstruction to that proof inference, not a counterexample to Liu’s theorem or to a later source version.

Two structural reductions can nevertheless be repaired. A compact moment-set argument gives a binary latent variable for the continuous candidate kernel. An exact-arithmetic certificate for the finite bound used in a conventional positive-semidefinite (PSD) argument, joined to an explicit signed-measure/operator bridge, gives separate concavity on each component’s two-moment slice

and hence component laws with at most three atoms each. We also prove a sharp entropy-zero asymptotic, certify one one-component face by 256-bit directed rounding, and eliminate the latent weight q exactly.

The generic reduced model has eleven variables and retains ten after eliminating q . The separately studied genuine 2×2 subface has seven variables and retains six. No theorem confines a global minimizer to that subface, and no global certificate is claimed.

Keywords: union-closed sets; Frankl’s conjecture; entropy method; moment sets; positive-semidefinite certificate; computer-assisted proof.

Claim-to-evidence matrix

Class	Result	Status and evidence
Analytically proved	Parameterised source-specific counterdirection; binary-latent minimizer; endpoint identity and uniform limit; exact q -elimination	Numbered results and proofs in this note
Exact computer-assisted	Finite-plus-tail arithmetic bound used by the PSD argument	Exact rational checker, fingerprints, and ten mutation tests; conventional analytic bridge stated separately
Directed-rounding computation	One-component atom-zero face	32,768 leaves at 256-bit precision, endpoint lemma, separate checker, and seven mutation tests
Conditional on the PSD bridge	At most three atoms in each component	Sequential two-moment-slice minimization
Open	Global $\Phi_C \geq 0$ on the full independently weighted $3 + 3$ -atom domain	Ten variables remain after eliminating q
Not established	$c = 153/400$ as a Frankl constant; any improved universal lower bound; a global 2×2 reduction	Explicit claim ceiling

Reduction map

$$\begin{aligned}
& \text{arbitrary continuous component laws} \\
& \downarrow \text{binary-latent extreme-point argument} \\
& (q, P_0, P_1) \\
& \downarrow \text{separate two-moment minimization, conditional on PSD} \tag{R} \\
& P_0, P_1 \text{ each supported on at most three points} \\
& \downarrow \text{parameter count} \\
& 11 \text{ variables} \longrightarrow 10 \text{ after exact } q\text{-elimination.}
\end{aligned}$$

Separately, on the genuine 2×2 subface,

$$7 \text{ variables} \longrightarrow 6 \text{ after exact } q\text{-elimination.} \tag{R2}$$

The second line is not a reduction of the first.

1. Frozen model and the unresolved obligation

All logarithms are natural. Changing the logarithm base multiplies every entropy term by the same positive constant and does not affect any inequality. Define binary entropy by

$$h(z) = -z \log z - (1 - z) \log(1 - z), \quad h(0) = h(1) = 0.$$

For $0 \leq s \leq 1$, set

$$K_s(x, y) = xy + s x(1 - x)y(1 - y), \quad x, y \in [0, 1].$$

For finite signed Borel measures σ, ρ on $[0, 1]$, define the bilinear form

$$H_s(\sigma, \rho) = \iint h(K_s(x, y)) \, d\sigma(x) d\rho(y),$$

and abbreviate $H_s(\sigma) = H_s(\sigma, \sigma)$ for any finite signed measure σ . This notation is used below in expressions such as $H_0(P_1 - P_0)$. Also put

$$D(P) = \int h(x) \, dP(x).$$

The frozen rational constants are

$$c = \frac{153}{400}, \quad m = 1 - c = \frac{247}{400}, \quad \tau = \frac{123}{500},$$

$$\beta = \frac{9501}{50000}, \quad \gamma = 1 - \beta = \frac{40499}{50000}, \quad C = 1 + 10^{-8} = \frac{100000001}{100000000}.$$

Let $q \in [0, 1]$, let P_0, P_1 be probability laws on $[0, 1]$, and write

$$M = (1 - q)P_0 + qP_1.$$

The numerator, denominator, and difference functional are

$$N(q, P_0, P_1) = \gamma H_0(M) + \beta \{(1 - q)H_\tau(P_0) + qH_\tau(P_1)\},$$

$$D(q, P_0, P_1) = D(M), \quad \Phi_C(q, P_0, P_1) = N(q, P_0, P_1) - CD(M).$$

The complement-variable mean condition is

$$\int x \, dM(x) \geq m. \tag{1.1}$$

The unresolved obligation is

$$\Phi_C(q, P_0, P_1) \geq 0 \tag{1.2}$$

for every feasible (q, P_0, P_1) . A uniform multiplier $C > 1$ is used because the entropy transfer to a union-closed family requires strict gain. An inequality with $C = 1$ alone would give no contradiction without an additional equality classification and limiting argument.

Current status

The following statements have been established within their stated scopes:

1. a source-mapped positive-curvature direction, valid for every $0 < \ell \leq 1$, for Liu's printed joint-concavity step;
2. a binary-latent reduction for the continuous kernel $h \circ K_\tau$;
3. the conditional-kernel PSD statement for $0 \leq \tau \leq 123/500$, supported by exact rational replay;
4. a sequential reduction to independently weighted component laws, each supported on at most three points;
5. a sharp entropy-zero asymptotic and its associated fixed-parameter ceiling;
6. a certified one-component face of the reduced domain; and
7. exact algebraic minimization in q on fixed component laws.

Statement (1.2) over the complete independently weighted $3 + 3$ -atom domain has not been established. After exact elimination of q , this generic domain is ten-dimensional before boundary degeneracies. Numerical searches did not locate a reliable negative point, but that observation is an exploratory record and is not used as proof evidence.

2. A parameterised positive-curvature direction in the arXiv-v1 proof domain

The source location is the proof of Theorem 12 in Liu's arXiv:2306.08824v1 ([2023, PDF p. 13](#)). After imposing the two aggregate constraints (76)–(77), the proof invokes Lemma 11 to treat objective (43) as concave on the joint feasible set and then derives the shared-weight representation (74)–(75). The checked source span is `main.tex:791–806`; the complete source-archive SHA-256 is `49ae8b3b25644223979f1706b81aa6d117775d8bcc0c2a03e1c3483bcb0c6ef0`.

Proposition 2.1 (source-to-model map). Let $S \in [0, 1]$ denote Liu's inclusion-probability variable and put $X = 1 - S$. In Example 5 and Theorem 12 of arXiv:2306.08824v1 choose

$$p(x) = 1 - x, \quad f_\ell(x) = \ell x p(x) = \ell x(1 - x), \quad 0 < \ell \leq 1. \quad (2.0)$$

This is protocol-valid because $0 \leq \ell x(1 - x) \leq \min\{x, 1 - x\}$. With $\tau = \ell^2$, the protocol probability in Liu's Example 5 becomes

$$\Pi_{s,t}(0, 0) = (1 - s)(1 - t) + f_\ell(1 - s)f_\ell(1 - t) = K_{\ell^2}(x, y),$$

where $x = 1 - s$ and $y = 1 - t$. Consequently Liu's objective (43), on the binary-latent representation with aggregate law $M = (1 - q)P_0 + qP_1$, is

$$(1 - \beta)H_0(M) + \beta\{(1 - q)H_{\ell^2}(P_0) + qH_{\ell^2}(P_1)\} - D(M). \quad (2.0a)$$

Replacing the final coefficient 1 by any real C does not change the curvature calculation below because the entire aggregate law remains fixed. Natural logarithms are used throughout; another

logarithm base multiplies the objective by a positive constant and leaves the curvature sign unchanged.

Source quantity	Complement-model quantity
inclusion variable S	$X = 1 - S$
$p(x) = 1 - x$, $f_\ell(x) = \ell x(1 - x)$	$K_{\ell^2}(x, y)$, with $\tau = \ell^2$
inclusion mean $c' = 3/8$	complement mean $1 - c' = 5/8$
aggregate constraint (76)	$\int x dM = 5/8$
aggregate constraint (77)	$\int f_\ell(x) dM = \ell/16$
one-measure slices in Lemma 11	fixed component mean and f_ℓ -moment

Theorem 2.2 (parameterised joint-concavity obstruction). Let

$$P = \frac{1}{2}\delta_1 + \frac{1}{4}\delta_0 + \frac{1}{4}\delta_{1/2}, \quad \nu = \delta_0 - \delta_{1/2},$$

and take

$$q = \frac{1}{2}, \quad P_0(t) = P + t\nu, \quad P_1(t) = P - t\nu, \quad |t| \leq \frac{1}{4}. \quad (2.1)$$

The measures in (2.1) are probability laws throughout the displayed interval, and

$$\frac{1}{2}P_0(t) + \frac{1}{2}P_1(t) = P. \quad (2.2)$$

The fixed aggregate moments are

$$\int x dP(x) = \frac{5}{8} = \frac{247}{400} + \frac{3}{400} > m, \quad \int f_\ell(x) dP(x) = \frac{\ell}{16}.$$

Thus the direction satisfies Liu's aggregate constraints at $c' = 3/8$ and $d = \ell/16$; in particular, $3/8 < 153/400$. The individual component moments are not fixed:

$$\begin{aligned} \int x dP_0(t) &= \frac{5}{8} - \frac{t}{2}, & \int x dP_1(t) &= \frac{5}{8} + \frac{t}{2}, \\ \int f_\ell dP_0(t) &= \frac{\ell}{16} - \frac{\ell t}{4}, & \int f_\ell dP_1(t) &= \frac{\ell}{16} + \frac{\ell t}{4}. \end{aligned} \quad (2.2a)$$

Equation (2.2a) is precisely why a one-measure concavity lemma on a fixed component slice cannot establish joint concavity on the larger aggregate-constrained pair space. Moreover,

$$\frac{1}{2}P_0(t)^{\otimes 2} + \frac{1}{2}P_1(t)^{\otimes 2} = P^{\otimes 2} + t^2\nu^{\otimes 2}.$$

Since $K_{\ell^2}(0, 0) = K_{\ell^2}(0, 1/2) = 0$ and

$$K_{\ell^2}(1/2, 1/2) = \frac{4 + \ell^2}{16},$$

bilinearity gives

$$H_{\ell^2}(\nu) = h\left(\frac{4 + \ell^2}{16}\right) > 0.$$

Every aggregate term, including $H_0(M)$, $D(M)$, the mean, and the aggregate f_ℓ -moment, is constant in t . Therefore, for every real C , if $\Phi_{C,\ell}$ denotes (2.0a) with $-D(M)$ replaced by $-CD(M)$, then

$$\Phi_{C,\ell}\left(\frac{1}{2}, P_0(t), P_1(t)\right) = \Phi_{C,\ell}\left(\frac{1}{2}, P, P\right) + \beta t^2 h\left(\frac{4 + \ell^2}{16}\right). \quad (2.3)$$

For every $0 < \ell \leq 1$, the second derivative along (2.1) is

$$2\beta h\left(\frac{4 + \ell^2}{16}\right) > 0. \quad (2.4)$$

At the frozen value $\ell = \sqrt{123/500}$, the entropy argument in (2.3) is 2123/8000. Equations (2.0)–(2.4) prove Proposition 2.1 and Theorem 2.2. Because the construction works for every $0 < \ell \leq 1$, it includes the “sufficiently small $\ell > 0$ ” regime stated in Liu’s Theorem 12 and invalidates the printed joint-concavity inference on the aggregate-constrained pair space.

The scope is narrow. The obstruction does not invalidate Liu’s one-measure Lemma 11, rule out a different proof of a shared-weight minimizer, affect Liu’s analytic non-explicit improvement, or remove the hypotheses from Liu’s conditional numerical value. The CISS 2024 proceedings version has not been inspected in full in this release, so no claim is made about whether that version contains the same inference; see `SOURCE_VERSION_MATRIX.md`.

3. Repaired cardinality reductions

3.1 From arbitrary latent mixing to a binary latent law

Liu’s Theorem 9 states a binary-latent reduction for a conditionally i.i.d. protocol. The result below is a self-contained reconstruction for the continuous kernel used here. Its contribution is to make attainment, continuity, the fixed-mean slice, and the extreme-point step explicit; no priority is claimed for the support number two.

Theorem 3.1 (binary-latent minimizer). For the continuous candidate kernel and a fixed aggregate-mean slice, the mixing-law objective attains a minimum at a law supported on at most two component measures. Equivalently, a minimizing conditionally independent and identically distributed (i.i.d.) coupling may be written as in (3.3).

For the original mean inequality, compactness first supplies a global minimizer with some realised mean $a \geq m$. The argument is then applied on the equality slice with mean a ; replacing the minimizer within that slice preserves feasibility for the original inequality.

Let $\mathcal{P} = \mathcal{P}([0, 1])$, equipped with the weak topology, and let

$$T(R) = R^{\otimes 2}.$$

The space $\mathcal{P}$ is compact and metrizable, and T is continuous. For a mixing law $\Lambda \in \mathcal{P}(\mathcal{P})$, define

$$B(\Lambda) = \int_{\mathcal{P}} R^{\otimes 2} d\Lambda(R). \quad (3.1)$$

The continuous affine image of $\mathcal{P}(\mathcal{P})$ under B is exactly the closed convex hull of the product laws. This representation keeps the full component law R . It does not replace R by its scalar mean.

On a slice of fixed aggregate mean, the conditionally-IID term and the denominator are affine in $B(\Lambda)$. The remaining nonlinear term is H_0 of the first marginal. If η is a finite signed direction of zero mean and $M_k(\eta) = \int x^k d\eta(x)$, the entropy series gives

$$H_0(\eta) = - \sum_{k \geq 2} \frac{M_k(\eta)^2}{k(k-1)} \leq 0. \quad (3.2)$$

Hence the objective is concave on each fixed-mean mixing-law slice. Its negative is a continuous convex functional, so the Bauer maximum principle gives a minimizing extreme point of the compact slice. This is a direct attainment-at-an-extreme-point argument; it does not require the stronger claim that every feasible mixing law has already been decomposed into binary feasible laws.

For completeness, let $g(R) = \int x dR(x)$. If an admissible mixing law Λ is not supported on at most two points of $\mathcal{P}$, choose three disjoint Borel sets A_i of positive Λ -mass. The three constants a_i , not all zero, can be chosen to solve the two homogeneous equations

$$\sum_{i=1}^3 a_i \Lambda(A_i) = 0, \quad \sum_{i=1}^3 a_i \int_{A_i} g(R) d\Lambda(R) = 0.$$

The signed measure with density a_i on A_i preserves mass and the fixed mean. For sufficiently small $\varepsilon > 0$, adding and subtracting ε times this measure gives two distinct admissible mixing laws whose midpoint is Λ . Such a Λ is not extreme. Therefore every extreme mixing law is supported on at most two component laws, and a minimizer has the form

$$B(\Lambda) = (1 - q)P_0^{\otimes 2} + qP_1^{\otimes 2}. \quad (3.3)$$

This proves Theorem 3.1. Continuity is essential to the attainment argument. It holds here because $h \circ K_\tau$ is continuous on $[0, 1]^2$. The argument does not, as stated, extend to every merely measurable conditionally-IID kernel.

3.2 Conditional-kernel PSD statement

The componentwise reduction uses the following statement.

Theorem 3.2 (conditional-kernel PSD statement). Let $0 \leq s \leq 123/500$, and let μ be a finite signed Borel measure on $[0, 1]$ satisfying

$$\int 1 d\mu = \int x d\mu = \int x^2 d\mu = 0. \quad (3.4)$$

Then

$$H_s(\mu) \leq 0. \quad (3.5)$$

Section 4 proves the analytic reduction and states the exact arithmetic certificate supporting (3.5), with their distinct trust boundaries.

3.3 Independently weighted three-atom components

Theorem 3.3 (independent three-atom reduction). Conditional on Theorem 3.2, the global minimum of Φ_C under (1.1) is attained by a triple in which each active component law has at most three support points. The two components have independent weights.

Put $f(x) = x(1 - x)$. For fixed a, b , consider the two-moment slice

$$\mathcal{S}(a, b) = \left\{ R \in \mathcal{P} : \int x dR = a, \quad \int f(x) dR = b \right\}. \quad (3.6)$$

Because $f(x) = x - x^2$, a signed difference of two members of $\mathcal{S}(a, b)$ has the three vanishing moments in (3.4). The slice is compact and convex. To see its support bound directly, suppose first that an atomic member has four distinct positive-weight atoms $x_1, \dots, x_4$. The 3×4 Vandermonde matrix with columns $(1, x_i, x_i^2)^\top$ has a nonzero null vector (v_i) . For small enough $\varepsilon > 0$, the weights $w_i \pm \varepsilon v_i$ remain nonnegative and preserve total mass, first moment, and second moment. The law is therefore not extreme; moving until a weight vanishes lowers the support. For a non-atomic law or a law with larger support, the same argument uses four disjoint positive-mass Borel sets and signed constant densities on those sets. Thus every extreme point has support at most three, in agreement with the moment-set theorems of Winkler (1988) and Pinelis (2016).

The full feasible set

$$[0, 1] \times \mathcal{P}([0, 1]) \times \mathcal{P}([0, 1])$$

is compact after imposing the closed mean condition (1.1). All kernels are bounded and continuous, so Φ_C is continuous and attains a global minimum. This use of the difference functional remains valid when $D = 0$.

Fix q, P_1 and the two moments of P_0 . For a signed direction η within the corresponding slice, the quadratic coefficient of $P_0 \mapsto \Phi_C(q, P_0, P_1)$ is

$$\gamma(1 - q)^2 H_0(\eta) + \beta(1 - q) H_\tau(\eta) \leq 0, \quad (3.7)$$

by (3.2) and (3.5). The $-CD$ term is affine. Thus the restriction is concave and may be minimized at an extreme point of the moment slice. After replacing P_0 , hold it fixed and repeat the argument for P_1 , whose quadratic coefficient is

$$\gamma q^2 H_0(\eta) + \beta q H_\tau(\eta) \leq 0. \quad (3.8)$$

This sequential operation preserves both components' relevant moments and the aggregate mean. It yields a global minimizer in which each component has at most three atoms. Their weight vectors are independent. At $q = 0$ or $q = 1$, the inactive component may be replaced by any point mass, so those closed faces are included without division by q or $1 - q$.

This proves Theorem 3.3. Before constraints and degeneracies, the corrected finite model has eleven scalar variables: two weights and three nodes in each component, plus q . After exact elimination of q , ten component variables remain. We order the nodes within each component as $0 \leq x_{i1} \leq x_{i2} \leq x_{i3} \leq 1$. Zero weights and coincident nodes are genuine closed faces and are not discarded or divided out.

The support bound itself is a standard moment-set phenomenon; see Winkler (1988) and Pinelis (2016). The point needing review here is the identification of the correct independent component

slices, the compatible quadratic form, and their sequential use, not the general moment-set principle. The common-weight three-pair conclusion in Liu's Theorem 12 is not treated as prior support for this independently weighted reduction because the printed joint-concavity inference used there is the obstruction in Section 2.

4. PSD argument, exact finite-bound certificate, and trust boundary

Theorem 4.1 (PSD/coercivity theorem). Let μ be a finite signed Borel measure on $[0, 1]$ with $\int 1 d\mu = \int x d\mu = \int x^2 d\mu = 0$. For every real $0 \leq s \leq 123/500$,

$$-H_s(\mu) \geq (1 - U) \sum_{k \geq 3} \frac{\left(\int x^k d\mu\right)^2}{k(k-1)} \geq 0, \quad (4.0)$$

where the exact rational certificate proves $U < 279521/500000 = 0.559042 < 1$. The theorem combines the conventional analytic bridge in Section 4.1 with the exact finite-plus-tail arithmetic in Section 4.2; the checker alone proves only the latter.

4.1 Analytic reduction

For μ satisfying (3.4), write

$$M_k = \int x^k d\mu(x), \quad z_k = \frac{M_k}{\sqrt{k(k-1)}} \quad (k \geq 3).$$

Since $|M_k| \leq \|\mu\|_{\text{TV}}$, the sequence z lies in ℓ^2 ; explicitly, $\sum_{k \geq 3} |z_k|^2 \leq \|\mu\|_{\text{TV}}^2/2$.

The two power series used below converge uniformly on their complete ranges. For $u = s(1-x)(1-y) \in [0, 123/500]$,

$$(1+u) \log(1+u) = u + \sum_{n \geq 2} \frac{(-1)^n u^n}{n(n-1)},$$

and, for $0 \leq K_s \leq 1$,

$$(1-K_s) \log(1-K_s) = -K_s + \sum_{m \geq 2} \frac{K_s^m}{m(m-1)}.$$

The second series is uniformly convergent even at $K_s = 1$ by the Weierstrass test. Because $|\mu| \otimes |\mu|$ is finite, uniform convergence permits termwise integration against the signed product measure. The three moment conditions remove the linear and logarithmic cross terms. Dropping only nonnegative square terms then gives

$$-H_s(\mu) \geq \sum_{k \geq 3} \frac{M_k^2}{k(k-1)} - \sum_{\substack{n \geq 3 \\ n \text{ odd}}} \frac{s^n}{n(n-1)} \left(\int x(1-x)^n d\mu(x) \right)^2. \quad (4.1)$$

For odd $n \geq 3$, the second square is a finite linear functional of z :

$$\int x(1-x)^n d\mu(x) = \sum_{k=2}^n (-1)^k \binom{n}{k} M_{k+1}. \quad (4.2)$$

Let $R_s : \ell^2(\{3, 4, \dots\}) \rightarrow \ell^2(\{3, 5, 7, \dots\})$ be the odd-row operator

$$(R_s z)_n = \sqrt{\frac{s^n}{n(n-1)}} \sum_{k=2}^n (-1)^k \binom{n}{k} \sqrt{k(k+1)} z_{k+1}, \quad n \geq 3 \text{ odd.} \quad (4.2a)$$

The trace estimate below first proves that this row map extends to a bounded operator. With that extension, (4.1) becomes

$$-H_s(\mu) \geq \|z\|_2^2 - \|R_s z\|_2^2. \quad (4.3)$$

The row Gram operator $R_s R_s^*$ on odd degrees has entries

$$G_{n,r}^{(s)} = \frac{s^{(n+r)/2} S(n, r)}{\sqrt{n(n-1)r(r-1)}},$$

where

$$S(n, r) = \sum_{k=2}^{\min(n,r)} \binom{n}{k} \binom{r}{k} k(k+1) = (nr + n + r - 1) \binom{n+r-2}{n-1} - 2nr. \quad (4.4)$$

The identity in (4.4) follows from two Vandermonde convolutions. The omitted row trace is bounded by

$$\text{tr}(G_{>N}^{(s)}) \leq \frac{2(4s)^{N+1}}{1-4s}, \quad (4.5)$$

where summing over every integer degree deliberately overcounts the odd rows. At the endpoint $s = \tau := 123/500$, $4s = 123/125 < 1$, so the infinite positive operator is trace class and the finite-row plus tail passage is legitimate. If $R_s^{(\leq N)}$ and $R_s^{(>N)}$ denote the retained and omitted output rows, their ranges occupy orthogonal output coordinates. Hence

$$R_s^* R_s = (R_s^{(\leq N)})^* R_s^{(\leq N)} + (R_s^{(>N)})^* R_s^{(>N)},$$

with no finite-tail cross term. This finite-plus-tail quantity bounds the operator norm $\|R_s\|^2 = \|R_s^* R_s\|$, which is precisely the negative square term in (4.3), rather than an unrelated proxy matrix. Section 4.2 certifies the quantity at $s = \tau$; entrywise monotonicity then covers every smaller s .

4.2 Exact arithmetic result

Set $s = \tau = 123/500$. The frozen calculation retains the 200 odd rows $3, 5, \dots, 401$. All Gram entries are nonnegative. Exact integer square-root enclosures turn their row sums into rational upper bounds, so Gershgorin's theorem gives

$$\|R_\tau^{(\leq 401)}\|^2 \leq 0.36808288510867215 \dots$$

The exact rational version of (4.5) gives

$$\|R_\tau^{(>401)}\|^2 \leq 0.1909587584662615 \dots$$

Their exact rational sum satisfies

$$U = 0.5590416435749337 \dots < \frac{279521}{500000} = 0.559042 < 1. \quad (4.6)$$

Combining (4.3) and (4.6) yields the stronger coercive inequality

$$-H_\tau(\mu) \geq (1 - U) \sum_{k \geq 3} \frac{M_k^2}{k(k-1)} \geq 0. \quad (4.7)$$

All finite entries and the tail bound increase with s , so the endpoint calculation transfers the same coercive bound to every $0 \leq s \leq \tau = 123/500$, proving (4.0) and completing Theorem 4.1.

4.3 What the replay establishes

The verifier in `psd_checker/` consumes the frozen JSON certificate and does not execute its producer. It recomputes every rational row sum and the tail, uses a combinatorial expression for (4.4) that is syntactically different from the producer's, checks that expression against 121 literal small-degree sums, and rejects ten controlled certificate mutations. The maximizing finite row is $n = 5$. The frozen certificate SHA-256 is `c17b72825047e2d252d252036afd06fa7830bdda903ef24f5c35a3440abd946a`.

This replay establishes an exact arithmetic certificate for the finite bound used in the analytic PSD argument, relative to the checker and frozen inputs. It is not a proof-assistant formalization. In particular, the entropy-series interchange, the signed-measure-to- ℓ^2 passage, the trace-class construction, and the Vandermonde identity remain conventional mathematics. The checker and analytic derivation were separately written in the same AI-assisted producer context; this is implementation diversity, not independent external verification, and neither has been verified in a foundational formal system.

5. Entropy-zero identity, sharp asymptotic, and ceiling

Let P be any probability law on $[0, 1]$, let $X \sim P$, and write

$$\mu = \mathbb{E}X, \quad D = D(P) = \mathbb{E}h(X), \quad p_k = \mathbb{E}X^k.$$

The aggregate IID term has the exact identity

$$\boxed{H_0(P) = 2\mu D - \sum_{k \geq 2} \frac{(\mu - p_k)^2}{k(k-1)}}. \quad (5.1)$$

To see the sign and normalization, write

$$-(1-x) \log(1-x) = x - \sum_{k \geq 2} \frac{x^k}{k(k-1)}$$

and apply Tonelli separately to the nonnegative series for X and for the independent product XY . The first entropy term contributes $2\mu \mathbb{E}[-X \log X]$; the displayed series supplies the remainder in (5.1).

Put

$$R(P) = \sum_{k \geq 2} \frac{(\mu - p_k)^2}{k(k-1)}.$$

A three-region split at $\delta = D^{1/4}$ gives the following law-independent modulus for $0 < D < 1/16$:

$$0 \leq \frac{R(P)}{D(P)} \leq \omega(D) := 3 \left(\frac{D}{L^2} + \frac{D^{1/2}}{L^2} + \frac{2}{L} \right), \quad L = \frac{1}{4} \log(1/D). \quad (5.2)$$

For orientation, write $b_k = \mu - p_k$ and split its expectation over $[0, \delta]$, $(\delta, 1 - \delta)$, and $[1 - \delta, 1]$. Entropy lower bounds give, respectively, D/L , $D/h(\delta)$, and $\min\{1, (k-1)t\}$, where $t \leq D/L$. Squaring these three contributions and summing against $1/[k(k-1)]$ yields (5.2). This is the step that makes the estimate uniform over the underlying law.

Thus $R(P) = o(D(P))$ uniformly over all laws on $[0, 1]$. On the exact zero-entropy face, $D(M) = 0$ if and only if the aggregate law is supported on $\{0, 1\}$. When $0 < q < 1$, this forces every positive-weight atom of both components into $\{0, 1\}$; at $q = 0$ or $q = 1$, only the active component is constrained. All kernels then take values in $\{0, 1\}$, so $N = D = 0$. The following statement controls every positive-entropy approach to that face, not merely the explicit family used for sharpness.

Because the conditionally-IID terms in N are nonnegative, (5.1)–(5.2) imply the following sharp endpoint result.

Liu’s Lemma 7 records the qualitative endpoint mechanism used to control laws approaching $\{0, 1\}$. The result here is a quantitative refinement: it supplies a law-uniform modulus, treats every feasible approach within the frozen functional, and proves sharpness by an explicit family. “Uniform” in Theorem 5.1 is global only over those entropy-zero approaches; it is not a global solution of the complete optimisation problem.

Theorem 5.1 (sharp uniform endpoint theorem). For every feasible sequence with $0 < D(M_n) \rightarrow 0$,

$$\liminf_{n \rightarrow \infty} \frac{N_n}{D(M_n)} \geq 2\gamma m = \frac{10003253}{10000000} = 1.0003253. \quad (5.3)$$

The constant is attained asymptotically by

$$P_0 = \delta_1, \quad P_1 = \delta_\varepsilon, \quad q = \frac{c}{1 - \varepsilon}, \quad \varepsilon \downarrow 0. \quad (5.4)$$

For $0 < \varepsilon \leq m$, (5.4) is feasible and has aggregate mean exactly m . If

$$k_\varepsilon = \varepsilon^2 + \tau \varepsilon^2 (1 - \varepsilon)^2,$$

then direct evaluation gives

$$\frac{N}{D} = \gamma \left\{ 2(1 - q) + q \frac{h(\varepsilon^2)}{h(\varepsilon)} \right\} + \beta \frac{h(k_\varepsilon)}{h(\varepsilon)}.$$

The two entropy ratios tend to zero and $q \rightarrow c$, which proves sharpness in (5.3). At the frozen C , the exact limiting margin is

$$2\gamma m - C = \frac{32529}{100000000} > 0. \quad (5.5)$$

The explicit modulus also proves $N > CD$ throughout the conservative punctured neighbourhood $0 < D \leq e^{-100000}$. This cutoff is not intended as a practical interval boundary; it records uniformity rather than only a pointwise limit.

The sharp family also imposes a ceiling on this frozen functional. If a new target c' is tested with the same β and C , feasibility near the endpoint requires

$$2\gamma(1 - c') \geq C.$$

Hence a necessary condition is

$$\boxed{c' \leq c_{\text{end}} = 1 - \frac{C}{2\gamma} = \frac{61995999}{161996000} = 0.38270080125435196 \dots} \quad (5.6)$$

Equality in (5.6) has zero leading endpoint slack. Positive endpoint slack requires strict inequality, and satisfying (5.6) is not sufficient for the global inequality. Even in the formal limit $C \downarrow 1$, the fixed β ceiling is

$$\frac{15499}{40499} = 0.3827008074273439 \dots \quad (5.7)$$

Equations (5.6)–(5.7) are mechanism-specific design constraints for the frozen functional, not upper bounds on all conditionally-IID methods.

6. Certified one-component atom-zero face

Theorem 6.1 (certified atom-zero face). On the complete face (6.1), $\Phi_C \geq 0$. The proof is analytic in p and near $x = 1$, and uses a 32,768-leaf, 256-bit MPFR directed-rounding certificate on the remaining compact interval.

The covered face is

$$q \in \{0, 1\}, \quad P = (1 - p)\delta_0 + p\delta_x, \quad px \geq m. \quad (6.1)$$

Put

$$k(x) = x^2\{1 + \tau(1 - x)^2\}, \quad A(x) = \gamma h(x^2) + \beta h(k(x)).$$

All pairs involving the atom at zero have zero kernel entropy, so

$$\Phi_C(p, x) = p^2 A(x) - Cph(x). \quad (6.2)$$

The entropy inequality

$$h(x^2) \geq xh(x) \quad (6.3)$$

follows, for example, by taking independent Bernoulli- x variables U, V and comparing $H(UV)$ with $H(UV \mid V) = xh(x)$. From (6.3), feasibility, and (5.5),

$$\partial_p \Phi_C(p, x) \geq (2\gamma m - C)h(x) > 0$$

for $x < 1$. The worst feasible weight is therefore $p = m/x$. It remains to prove

$$J(x) := m\{\gamma h(x^2) + \beta h(k(x))\} - Cxh(x) \geq 0, \quad m \leq x \leq 1. \quad (6.4)$$

The interval certificate partitions $[m, 9999/10000]$ into 32,768 rational leaves and evaluates a monotonicity and concavity enclosure for (6.4) using 256-bit MPFR directed rounding. A separately implemented verifier recomputes every leaf. The replayed minimum lower bound is

$$4.75628699253959553830 \times 10^{-5} > 0.$$

An analytic endpoint substitution $y = 1 - x$ supplies an affine lower bound in $-\log y$ on $[9999/10000, 1)$, and $J(1) = 0$ exactly. The endpoint slope lower bound replayed as

$$2.349353535011900000000 \times 10^{-1} > 0.$$

This proves Theorem 6.1. The producer and verifier share MPFR 4.2.2, the compiler toolchain, and the certificate's endpoint operation order. The replay is therefore a strong implementation check, but not an independent arithmetic library or a formal proof.

7. Exact elimination of q : ten generic variables and a six-variable subface

Proposition 7.1 (exact elimination of the latent weight). For fixed component laws, the feasible q -set is a closed interval and the minimum of Φ_C is obtained by the endpoint/vertex rule below. Consequently the generic $3 + 3$ -atom model has ten variables after eliminating q , while the genuine 2×2 subface has six.

Fix P_0, P_1 and define

$$G_{ij} = H_0(P_i, P_j), \quad T_i = H_\tau(P_i), \quad d_i = D(P_i), \quad \mu_i = \int x dP_i(x).$$

Set

$$F_i = \gamma G_{ii} + \beta T_i - C d_i,$$

and

$$A = G_{00} + G_{11} - 2G_{01} = H_0(P_1 - P_0).$$

Direct bilinear expansion gives the exact identity

$$\boxed{\Phi_C(q) = (1 - q)F_0 + qF_1 - \gamma Aq(1 - q).} \quad (7.1)$$

Write

$$\Phi_C(q) = aq^2 + bq + d, \quad a = \gamma A, \quad b = F_1 - F_0 - a, \quad d = F_0. \quad (7.2)$$

The mean condition is independently affine:

$$\mu_0 + q(\mu_1 - \mu_0) \geq m. \quad (7.3)$$

Thus the feasible values of q form an empty set or one closed interval $[L, U]$. Explicitly, intersect $[0, 1]$ with $[(m - \mu_0)/(\mu_1 - \mu_0), \infty)$ if $\mu_1 > \mu_0$, or with $(-\infty, (m - \mu_0)/(\mu_1 - \mu_0)]$ if $\mu_1 < \mu_0$. If $\mu_1 = \mu_0$, the interval is $[0, 1]$ when $\mu_0 \geq m$ and empty otherwise. The minimum in q occurs at L , at U , or, when $a > 0$, at the vertex

$$q_* = -\frac{b}{2a} = \frac{1}{2} - \frac{F_1 - F_0}{2\gamma A}$$

if that vertex lies in (L, U) . A rigorous checker can avoid division: the interior condition is

$$-2aU < b < -2aL,$$

and the vertex value is nonnegative exactly when

$$4ad - b^2 \geq 0. \quad (7.4)$$

When $a = 0$, the objective is affine and only feasible endpoints are needed. The singular branches $a = 0$ and $\mu_0 = \mu_1$ must be kept separately. Both signs of A occur, so universal concavity in q is unavailable.

A generic three-atom probability law has five continuous parameters, two free weights and three ordered nodes. The generic $3 + 3$ -atom model therefore has $5 + 5 + 1 = 11$ variables before Proposition 7.1 and ten after eliminating q .

A genuine two-atom probability law has three continuous parameters, one weight and two ordered nodes. Therefore a genuine 2×2 branch has seven variables before (7.1): three for each component and q . Exact elimination of q leaves six component variables. On a mean-crossing branch, equation (7.3) determines the same eliminated q ; it does not remove a second component variable. Mean equality is also not universal: exact slack-mean minimizers exist on the entropy-zero face. A retained 80-decimal diagnostic also places a positive-entropy fixed-pair minimum at a slack-mean endpoint, but that latter observation is not interval-certified.

No theorem currently confines a global minimizer to the 2×2 face. Nor does a theorem exclude every feasible interior vertex, supply a second independent support or stationarity identity, or remove a genuine component parameter on every six-variable branch of that face. The unresolved global problem is therefore the generic ten-variable reduced domain; a six-dimensional obstruction already appears on the studied surface.

8. Assurance boundary and artifact map

The completed statements in this note combine conventional proofs with exact or directed-rounding computation. They have undergone local and cross-platform same-source arithmetic replay and adversarial checking. They have not been formalized in a proof assistant, independently reimplemented, or reviewed by an independent specialist. Most importantly, they do not imply (1.2) on the complete domain and therefore do not establish 153/400 as a valid Frankl constant.

8.1 Position relative to earlier results

Result here	Closest earlier result	Bounded contribution claimed here
Proposition 2.1 and Theorem 2.2	Liu, Theorem 12 and Example 5	Exact source map and a parameterised counterdirection against the printed joint-concavity inference
Theorem 3.1	Liu, Theorem 9	Self-contained continuous-kernel reconstruction with explicit attainment and slice argument; no priority claim for binary support
Theorem 3.3	Winkler and Pinelis moment-set theorems	Standard support theory applied on the correct independent component slices using the compatible quadratic form
Theorem 4.1	Liu, Lemma 11	Conventional infinite-series/operator proof plus an exact-arithmetic finite-bound certificate; priority remains bounded uncertainty
Theorem 5.1	Liu, Lemma 7	Law-uniform quantitative modulus and sharpness within the frozen functional
Proposition 7.1	Elementary quadratic minimisation	Explicit branch-safe q -elimination and honest residual dimension count

The contrast between the blocked and repaired support routes is:

Printed shared-weight route	Repaired independent-weight route
aggregate constraints on (P_0, P_1)	fix each component's own two moments
invoke one-measure slice concavity jointly	apply separate concavity sequentially
common three weights for paired atoms	at most three atoms per component with independent weights
nine-dimensional numerical programme	eleven variables, ten after exact q -elimination

8.2 Theorem-level assurance matrix

Claim	Analytic dependency	Computational dependency	Assurance and failure effect
Proposition 2.1 / Theorem 2.2	Liu v1 source, direct moment and curvature calculation	none	Conventional source-specific proof; failure affects the central obstruction
Theorem 3.1	compactness, continuity, Bauer/extreme-point argument	none	Conventional proof; failure affects only the binary-latent repair
Theorems 3.2 and 4.1	entropy series, signed measures, operator and tail bounds	exact rational certificate and ten mutations	Exact arithmetic within a conventional semantic bridge; failure affects Theorem 3.3
Theorem 3.3	Theorem 4.1, moment-set geometry, sequential minimisation	PSD replay above	Conditional on the PSD bridge; failure affects the independent-support repair
Theorem 5.1	identity (5.1), uniform modulus, sharp family	diagnostics are non-deductive	Conventional proof; failure affects the endpoint ceiling, not Theorem 2.2
Theorem 6.1	analytic reduction and endpoint lemma	32,768-leaf MPFR certificate and seven mutations	Directed-rounding certificate; failure affects only the certified face
Proposition 7.1	quadratic algebra and affine feasibility	samples are illustrative only	Conventional proof; failure affects the dimension count
Global inequality (1.2)	all preceding reductions plus a missing global certificate	none exists	Open; no Frankl bound follows

The companion files intended for inspection are:

- `proof_repair/ANALYTIC_PROOF_REPAIR.md`: complete binary-latent, counterdirection, PSD bridge, and independent-support arguments;
- `psd_checker/MATHEMATICAL_DERIVATION.md` and `psd_checker/`: separately written certificate derivation, verifier, fingerprints, negative controls, and receipts;
- `endpoint_neighborhood/ENDPOINT_THEOREM.md`: full identity, uniform modulus, sharpness family, and finite punctured neighbourhood;
- `interval_pilot/ANALYTIC_REDUCTION.md` and `interval_pilot/`: the certified one-component face;
- `dimension_reduction/REPORT.md`: exact q -elimination and the six-dimensional 2×2 subface; the generic ten-variable count is derived in Proposition 7.1 and summarized in `REDUCTION_MAP.md`; and

- `MANIFEST.sha256` and `verify_candidate.py`: package-level file binding and replay entry point.

The full theorem-to-artifact routing table is in `THEOREM_DEPENDENCY_MAP.md`; focused questions for future specialist review are in `REVIEWER_QUESTIONS.md`. These dossier files are not additional proof evidence.

9. Limitations, availability, and declarations

Limitations. The global ten-variable inequality is open; the paper proves no new Frankl bound. The CISS 2024 full text has not been assessed. The literature search is bounded rather than exhaustive. Exact and interval replays are same-source checks and do not establish independent mathematical verification, specialist review, editorial peer review, or priority.

Data and code availability. All manuscript sources, analytic supplements, checkers, frozen certificates, receipts, negative controls, environment specifications, and the manifest are included in the versioned public research package at <https://github.com/ipitchford/frankl-concavity-obstruction> and archived under DOI [10.5281/zenodo.21938497](https://doi.org/10.5281/zenodo.21938497).

Ethics. The work uses no human participants, animals, personal data, or restricted datasets; institutional ethics review was not applicable.

Author contributions. Anonymous: conceptualisation, investigation, methodology, validation, writing, and responsibility for the release decision. AI-assisted tools supported derivation reconstruction, code, literature search, drafting, and producer-side checking; they are not authors.

Funding. No external funding is declared.

Competing interests. None declared.

AI assistance disclosure. Automated systems contributed substantially to mathematical exploration, implementation, checking, and editing. Their outputs were subjected to the bounded gates recorded in the package. Model review and same-source replay are not represented as independent specialist or peer review.

References

- Cambie, S. (2025). *Better bounds for the union-closed sets conjecture using the entropy approach* (arXiv:2212.12500v2). arXiv. <https://arxiv.org/abs/2212.12500v2>
- Liu, J. (2023). *Improving the lower bound for the union-closed sets conjecture via conditionally IID coupling* (arXiv:2306.08824v1). arXiv. <https://arxiv.org/abs/2306.08824v1>
- Liu, J. (2024). Improving the lower bound for the union-closed sets conjecture via conditionally IID coupling. In *2024 58th Annual Conference on Information Sciences and Systems (CISS)* (pp. 7–12). IEEE. <https://doi.org/10.1109/CISS59072.2024.10480167>
- Pinelis, I. (2016). On the extreme points of moments sets. *Mathematical Methods of Operations Research*, 83(3), 325–349. <https://doi.org/10.1007/s00186-015-0530-0>
- Winkler, G. (1988). Extreme points of moment sets. *Mathematics of Operations Research*, 13(4), 581–587. <https://doi.org/10.1287/moor.13.4.581>
- Yu, L. (2023). Dimension-free bounds for the union-closed sets conjecture. *Entropy*, 25(5), Article 767. <https://doi.org/10.3390/e25050767>

The complete status-oriented bibliography is in `REFERENCES.md`.